

Best Time for Vacationing

Anshika Garg, Piter Garcia, Wander Bravo

1. DESCRIPTION

The aim of this project is to learn the Big Data and Data Mining techniques to implement them in a project chosen by our team. The name of the project is Best Time for Vacationing, and we will be using the weather dataset of the United States and other countries such as Ecuador, the Dominican Republic, and India to analyze, compare, and show the weather trends in particular areas. We will make use of the Data Mining techniques to approximate future climates in these particular areas through their dataset. Thus, we will be able to predict places with a climate that best suits the input temperature of users. Furthermore, we are planning to design an interface to help users create a list of the best places for them to vacation as per their desired input temperature and behavior.

The datasets consist of the temperature, humidity, and precipitation of some locations. These datasets have been collecting weather information for the past fifty-four years. This information will help us analyze the trends of some areas over the past years, making our results more reliable. Our main goal is to make querying for the weather information easy and reliable. This can be accomplished through our software application by taking the input of users and providing them with a graphical output that can be easily interpreted.

To achieve the objective of our project we are going to make use of tools such as PHP, HTML and MySQL. In fact, we plan to use data cleaning-techniques to clean all datasets downloaded from the Internet and convert them into a useable database with MySQL. This process would allow us to remove all unwanted information to make the data more practical for the application while avoiding delays.

Once we have achieved the criteria mentioned above, we will proceed to expand our database by adding the datasets of other countries. These countries include Ecuador, the Dominican Republic, and India, whose datasets will be used to implement the above strategy to provide users with a diversity of locations that best fit their requirements. In addition, we will attempt to develop an extra feature for

our application. This feature is based on users previous selected locations to prioritize the list of suggested places. The suggested places are chosen comparing locations that were previously selected with those that have a similar climate.

2. CURRENT STATUS

Currently, we have started to gather climate information from the United States for our database. We were able to obtain temperature information, from all states, which has been collected since 1763. The task of collecting datasets from the United States has been so time consuming, it has precluded us from gathering weather datasets from Ecuador, the Dominican Republic, and India at this present time. However, we will proceed to collect the necessary weather information from the countries mentioned above to obtain their datasets and include them in our database. Finally, all sample datasets that have been downloaded will be cleaned to remove the unwanted information, which will then convert the data into usable SQL database.

3. TABLES

The tables below contain the datasets that we have obtained. However, they only represent a summary of the temperatures from some of the stations in the United States. It is important to highlight that each one of the station, or even a summary of all stations, contain huge data. Due to the large size of the data obtained by stations that are gathering weather information, it would be impossible for us to attach their respective datasets in the project.

4. CITATIONS/REFERENCES

<http://www.ncdc.noaa.gov/cdo-web/>
<http://www.ncdc.noaa.gov/cdo-web/datasets>

Figure 1: Annually Dataset Example.

STATION	STATION_NAME	ELEVATION	LATITUDE	LONGITUDE	DATE	TPCP	EMXT	EMNT
COOP:310301	ASHEVILLE NC US	682.1	35.5954	-82.5568	201001	413	59	10
COOP:310301	ASHEVILLE NC US	682.1	35.5954	-82.5568	201002	328	61	17
COOP:310301	ASHEVILLE NC US	682.1	35.5954	-82.5568	201003	286	73	25
COOP:310301	ASHEVILLE NC US	682.1	35.5954	-82.5568	201004	149	89	37
COOP:310301	ASHEVILLE NC US	682.1	35.5954	-82.5568	201005	265	85	40
COOP:310301	ASHEVILLE NC US	682.1	35.5954	-82.5568	201006	272	95	57
COOP:310301	ASHEVILLE NC US	682.1	35.5954	-82.5568	201007	229	95	62
COOP:310301	ASHEVILLE NC US	682.1	35.5954	-82.5568	201008	294	95	60
COOP:310301	ASHEVILLE NC US	682.1	35.59528	-82.55667	201009	325	90	48
COOP:310301	ASHEVILLE NC US	682.1	35.59528	-82.55667	201010	99	82	33
COOP:310301	ASHEVILLE NC US	682.1	35.59528	-82.55667	201011	348	72	23
COOP:310301	ASHEVILLE NC US	682.1	35.59528	-82.55667	201012	156	55	8

Figure 2: Monthly Dataset Example

STATION	STATION_NAME	ELEVATION	LATITUDE	LONGITUDE	DATE	TPCP	EMXT	EMNT
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100101	40	39	-328
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100201	208	-39	-289
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100301	67	222	-189
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100401	382	228	-33
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100501	1020	328	-17
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100601	881	278	78
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100701	496	317	111
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100801	488	339	61
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100901	1045	233	-6
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20101001	569	278	-56
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20101101	55	161	-233
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20101201	53	0	-256

Figure 3: Daily Dataset Example

STATION	STATION_NAME	ELEVATION	LATITUDE	LONGITUDE	DATE	TMAX	TMIN	PRCP
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100101	-178	-311	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100102	-244	-322	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100103	-194	-289	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100104	-167	-200	15
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100105	-133	-167	9999
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100106	-133	-172	9999
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100107	-150	-278	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100108	-233	-328	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100109	-233	-322	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100110	-117	-244	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100111	-67	-128	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100112	-78	-122	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100113	-17	-89	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100114	39	-72	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100115	-67	-72	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100116	22	-50	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100117	33	-44	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100118	6	-172	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100119	-56	-183	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100120	-67	-139	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100121	-67	-94	25
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100122	-44	-67	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100123	-6	-44	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100124	0	-11	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100125	-11	-161	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100126	-161	-233	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100127	-167	-222	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100128	-167	-283	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100129	-189	-283	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100130	-156	-267	0
GHCND:USC00327027	PETERSBURG 2 N ND US	466.3	48.0355	-98.01	20100131	-150	-272	0